

LINE FOLLOWER

Autonomous line-tracking robot challenge

Team Size	1-3 members	Entry Fee	Rs. 200
Prize	To be announced	Venue	To be announced

Objective

Build a fully autonomous robot that senses and follows a marked line (black line on white background, or vice versa) from start to finish in the shortest possible time.

Robot Specifications

- Maximum robot dimensions: 25 cm x 25 cm x 25 cm.
- Robot must be fully autonomous - no manual or remote control of any kind is permitted once the run starts.
- Wireless communication that assists navigation is not allowed.
- On-board battery power only.

Track & Gameplay

- Track will contain curves, sharp turns, and may include one or more break/discontinuity sections and a ramp, revealed on the event day.
- A run ends when the robot completes the track, goes off-line beyond the permitted recovery distance, or the time limit expires.
- Each team is allowed a fixed number of attempts (typically two to three); the best valid run counts.
- Manually lifting and replacing the robot after going off-track is allowed a limited number of times, each with a time penalty.

General Conduct

- Participants must carry a valid college ID card at all times during the event.
- Any form of misconduct, use of abusive language, or unsporting behaviour will lead to immediate disqualification.
- Decisions of the judges and organizing committee are final and binding.
- The organizers are not responsible for any damage, loss, or injury during the event; participants take part at their own risk.
- Participants must report to the venue at least 30 minutes before their scheduled slot.

Safety Guidelines

- Sharp tools, open flames, lasers, and hazardous chemicals are strictly prohibited unless pre-approved by the organizers.
- All electrical connections must be properly insulated; loose wiring may lead to disqualification on safety grounds.
- Batteries must be handled with care; short-circuits or overheating incidents must be reported immediately.

- Safety gear (gloves/goggles, where applicable) is recommended and may be mandatory for certain sub-tasks.

Disqualification Criteria

- Missing the reporting or check-in deadline for your slot.
- Using a robot/device or submission that does not meet the size, weight, or component specifications above.
- Remote interference with another team's device or run.
- Plagiarism or use of an undisclosed pre-built solution where original work was required.
- Any attempt to bribe, threaten, or unduly influence judges or coordinators.

These rules are provisional and general in nature. The organizing committee may issue clarifications, additional rounds, or minor changes closer to the event date; the latest version will always be posted on the PLASMA 2K26 website.